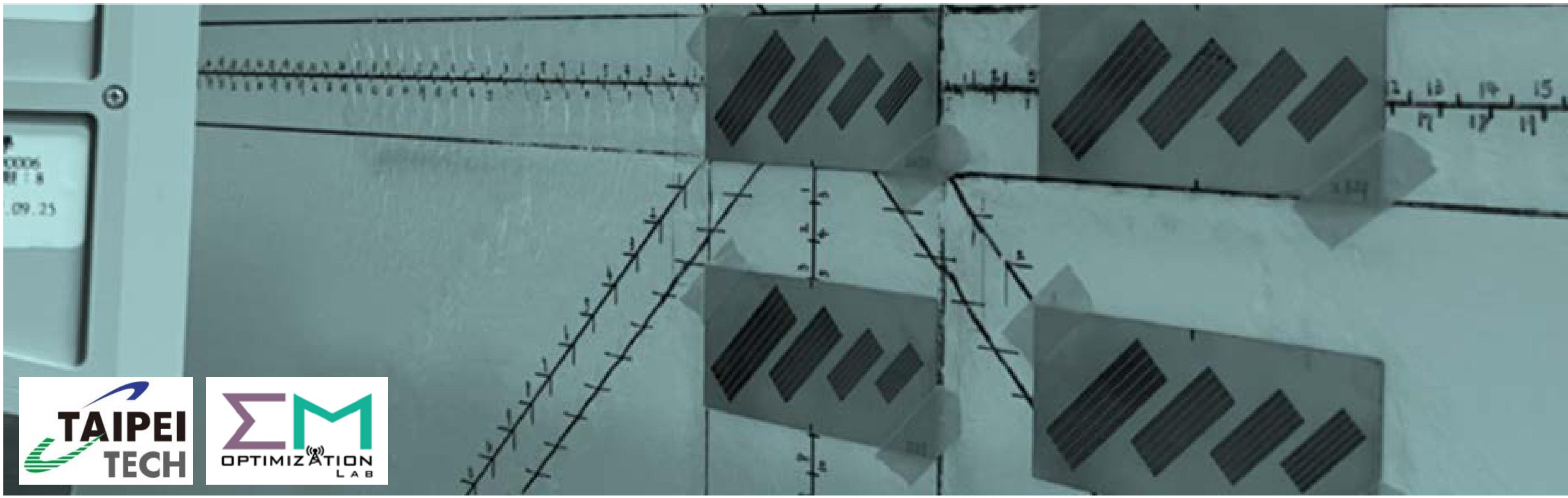
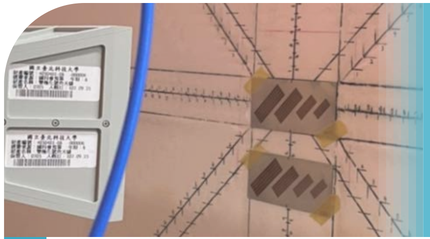
Linear Algebra
線性代數



Chapter 3: Orthogonality

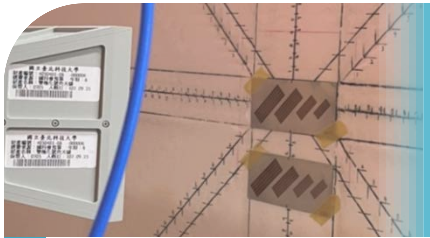
Yen-Sheng Chen (陳晏笙)

OCW Course Content



Agenda

1. Orthogonality of Subspaces
2. Projection onto 1-D Subspaces
3. Least Squares Approximations
4. Orthonormal Bases
5. Gram-Schmidt Process



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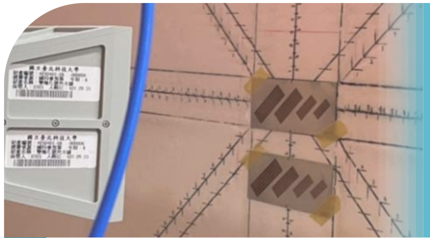


Roadmap of Chapters 2 & 3

Chapter 2

From individual vectors to a set of vectors: ____________

Cope with $\mathbf{A}\mathrm{x} = \mathrm{b}$ for the existence and uniqueness of x

Understand the terms to describe a space: ____________

Four essential subspaces of a matrix

Application 1: Graph

Application 2: “Matrixized”

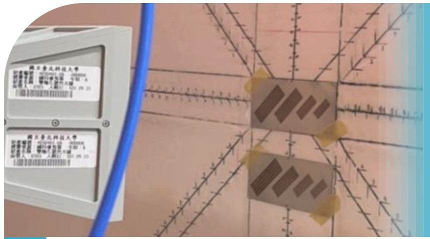
Chapter 3

Two special bases are the best; this chapter details the first one: ____________

Cope with $\mathbf{A}\mathrm{x} = \mathrm{b}$ for the unsolvable situation: ____________

How do we develop a better basis?

Application: Hilbert space



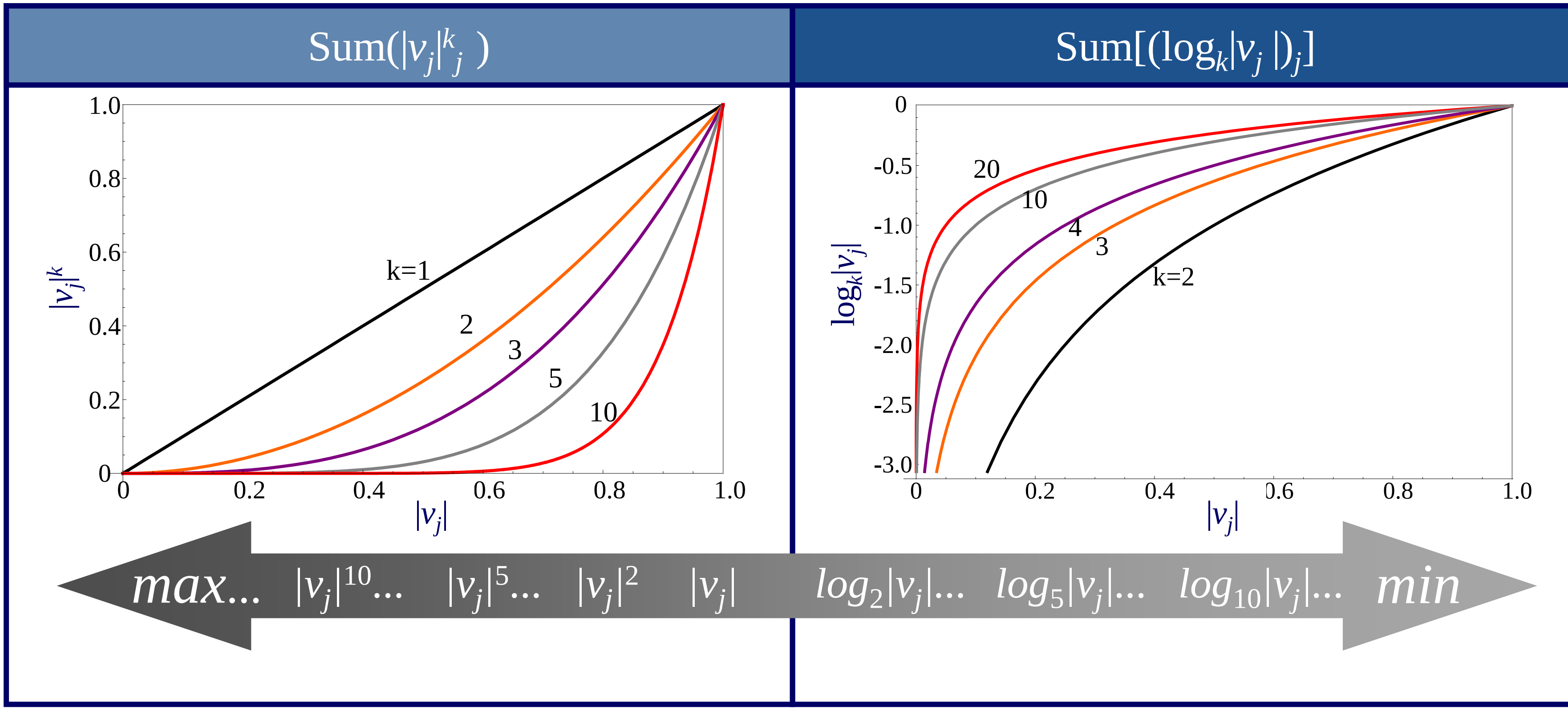
Norm of a Vector

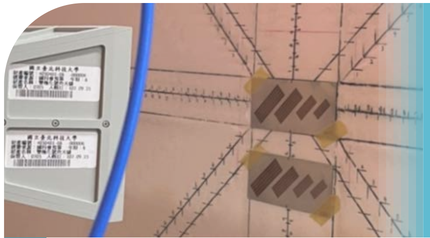
$$\mathrm{v} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}^T$$

- Norm: the measure for the length of a vector

l^1 norm	$\lVert \mathrm{v} \rVert_1 = \lvert v_1 \rvert + \lvert v_2 \rvert + \cdots + \lvert v_n \rvert$
l^2 norm	$\lVert \mathrm{v} \rVert_2 = \sqrt{\lvert v_1 \rvert^2 + \lvert v_2 \rvert^2 + \cdots + \lvert v_n \rvert^2} = \sqrt{\mathrm{v}^T \mathrm{v}}$
l^P norm	$\lVert \mathrm{v} \rVert_p = \left(\lvert v_1 \rvert^p + \lvert v_2 \rvert^p + \cdots + \lvert v_n \rvert^p \right)^{\frac{1}{p}}$
l^∞ norm	$\lVert \mathrm{v} \rVert_\infty = \max\left(\lvert v_1 \rvert, \lvert v_2 \rvert, \cdots, \lvert v_n \rvert \right)$
l^0 norm	$\lVert \mathrm{v} \rVert_0 =$ the number of non-zero component

Rules of Performance Index



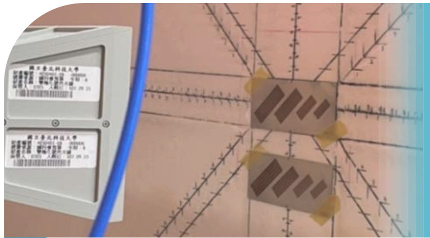


Norm of a Matrix

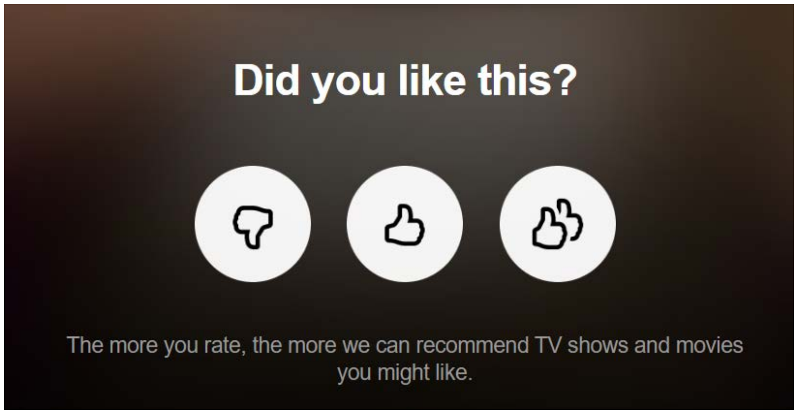
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

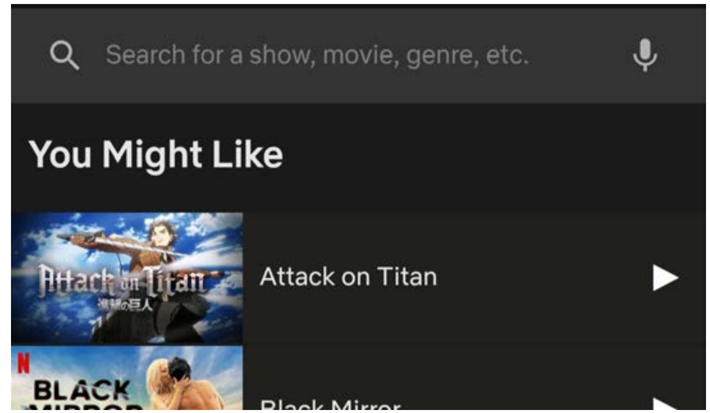
- A matrix also can be defined with a norm, which measures the size
- σ_i: the i^{th} singular value

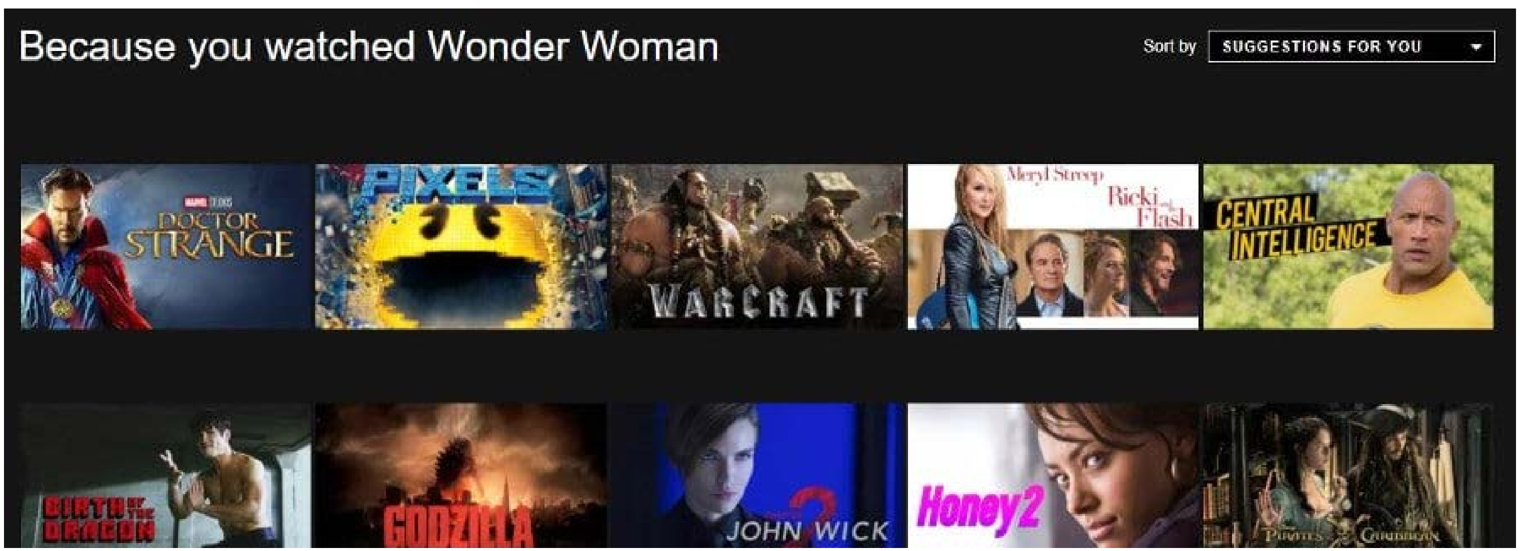
$\|\mathbf{A}\|_2$ norm	$\left\|\mathbf{A}\right\|_2 = \sigma_1$
$\|\mathbf{A}\|_F$ norm (Frobenius norm)	$\left\|\mathbf{A}\right\|_F = \sqrt{\left\|a_{11}\right\|^2 + \left\|a_{12}\right\|^2 + \cdots + \left\|a_{mn}\right\|^2}$
$\|\mathbf{A}\|_N$ norm (Nuclear norm)	$\left\|\mathbf{A}\right\|_N = \sigma_1 + \sigma_2 + \cdots \sigma_r$

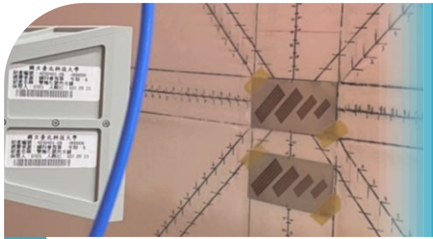


Matrix Completion Problem







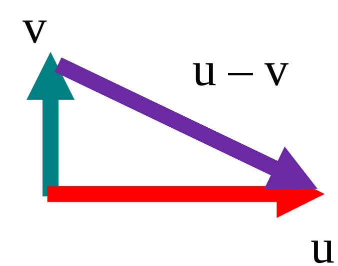


Orthogonal Vectors

$$\mathrm{v} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}^T$$

- l^2 norm can be expressed as the squared root of $\mathrm{v}^T\mathrm{v}$
- Example: $\mathrm{v} = [1 \ \ 1 \ \ 6 \ \ 6]^T$
- Pythagoras formula: u and v are perpendicular if

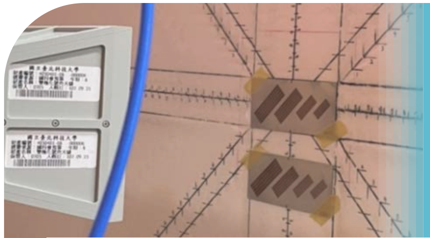
$$\|\mathrm{u}\|^2 + \|\mathrm{v}\|^2 = \|\mathrm{u} - \mathrm{v}\|^2$$



⇒ u and v are orthogonal if

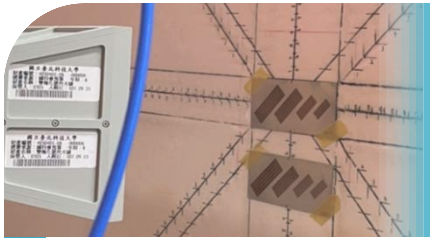
$$\mathrm{u}^T\mathrm{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \sum_{i=1}^{n} u_i v_i = 0$$

$\mathrm{u}^T\mathrm{v}$: inner product of u and v



Why Orthogonal Vectors are Important?

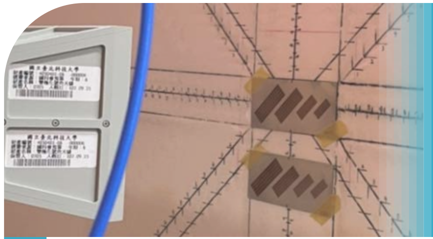
- If non-zero vectors $v_1, v_2, \ldots, v_k$ are mutually orthogonal, they must be linearly independent
- Suppose they are dependent → $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k = 0$ → but $v_1^T(c_1 v_1 + c_2 v_2 + \ldots + c_k v_k) = c_1 v_1^T v_1 = 0$ → c_1 must be 0



Orthogonal Subspaces

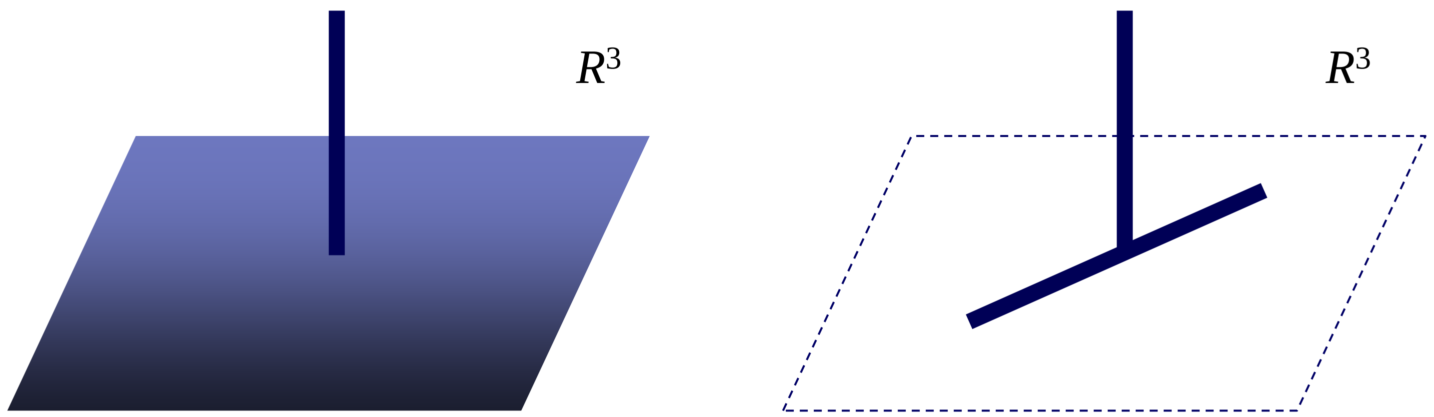
- Two subspaces V and W of the same space R^n are orthogonal if ***<u>every</u>*** vector v in V is orthogonal to ***<u>every</u>*** vector w in W
- That is, $\mathrm{v}^T\mathrm{w} = 0$ for all v and w

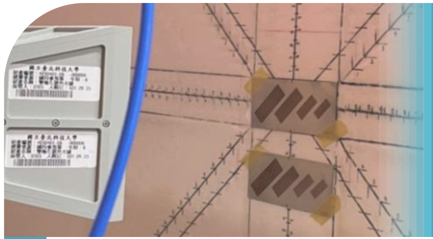
- Example:
  - V is a plane spanned by $\mathrm{v}_1 = [2\ \ 1\ \ 0\ \ 0]^T$ and $\mathrm{v}_2 = [-3\ \ 6\ \ 0\ \ 0]^T$, and W is another plane spanned by $\mathrm{w}_1 = [0\ \ 0\ \ 4\ \ 7]^T$ and $\mathrm{w}_2 = [0\ \ 0\ \ -2\ \ -1]^T$
  - How do we express all the vectors in V and those in W?
  - Why are the 2 subspaces orthogonal?
  - How do we identify two orthogonal subspaces efficiently?
  - Can two planes in R^3 be orthogonal?



Orthogonal Complement

- Given a subspace V of R^n, the space formed by all vectors orthogonal to V is called the orthogonal complement of V
- Denoted by $V^\perp$
- V and W can be orthogonal without being complement
- If $W = V^\perp$, then $V = W^\perp$; $V^{\perp\perp} = V$
- Example:



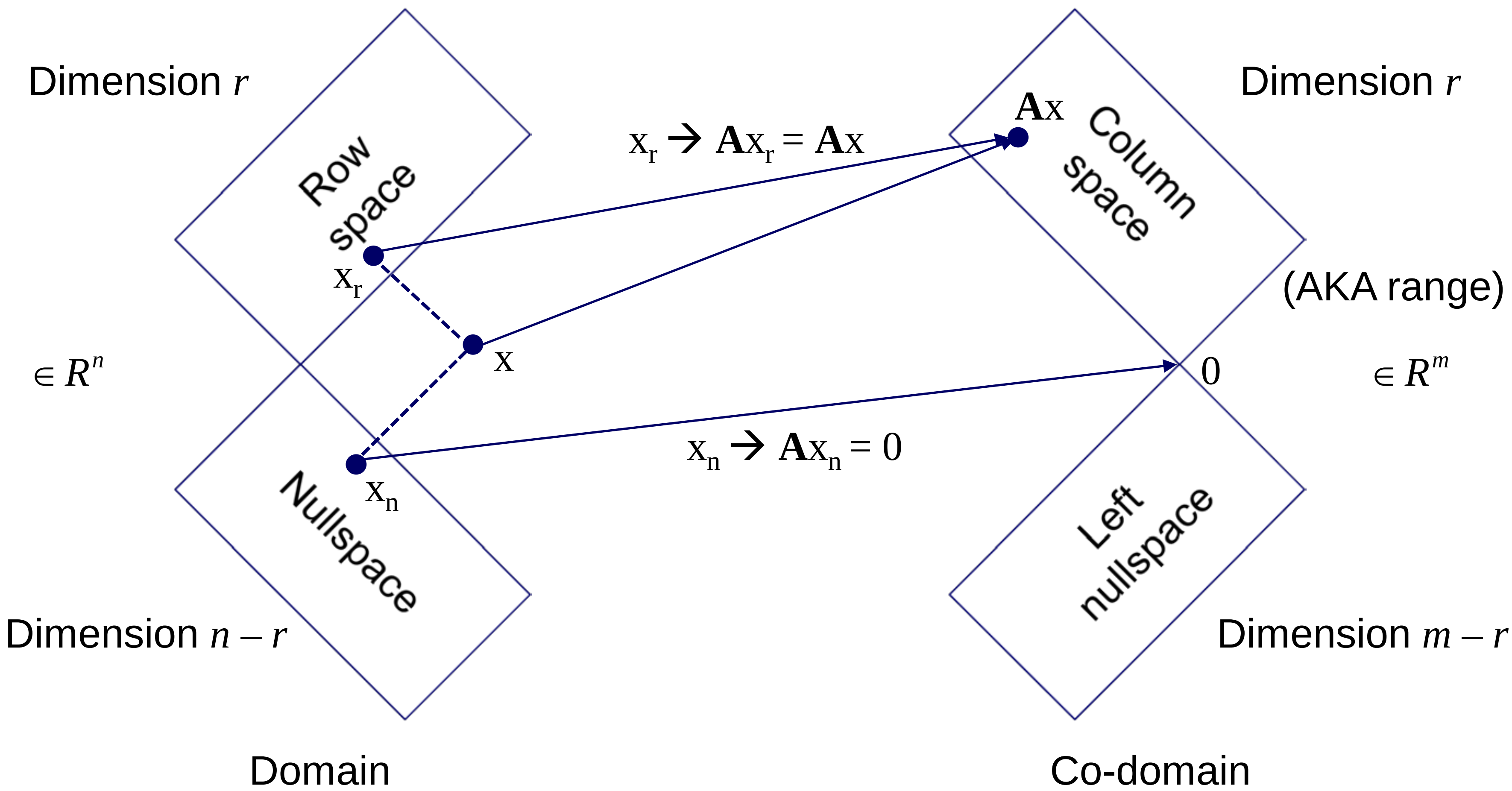


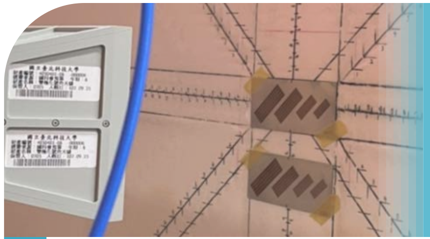
Fundamental Theorem of Linear Algebra, Part 2

$\mathrm{N}(\mathbf{A}) = \mathrm{C}(\mathbf{A}^T)^{\perp}$, since $r + (n - r) = n$

$\mathrm{N}(\mathbf{A}^T) = \mathrm{C}(\mathbf{A})^{\perp}$, since $r + (m - r) = m$

Transformation of A

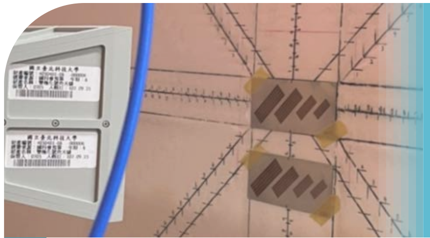




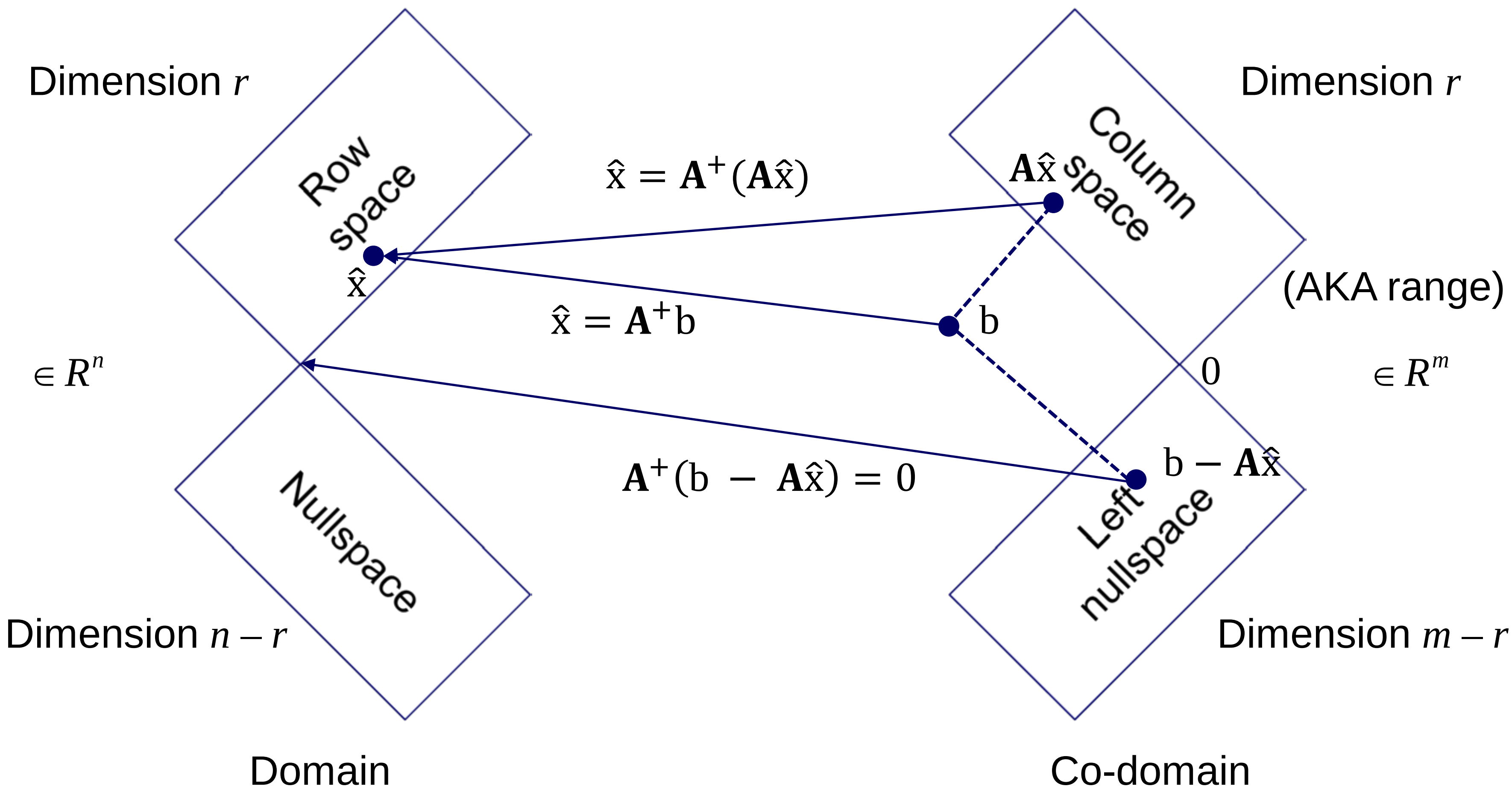
An Example

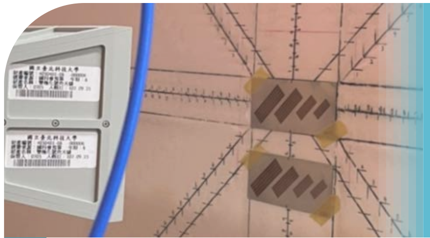
For a system $\mathbf{A}\mathrm{x} = \mathrm{b}$ where $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}$, $\mathrm{x} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$, and $\mathrm{b} = \begin{bmatrix} 9 \\ 18 \end{bmatrix}$,

- find the bases for the row space and the nullspace of $\mathbf{A}$
- Split x into a row-space component x_r and a nullspace component x_n
- Find the pseudoinverse of $\mathbf{A}$
- Use the pseudoinverse to recover the row-space component x_r



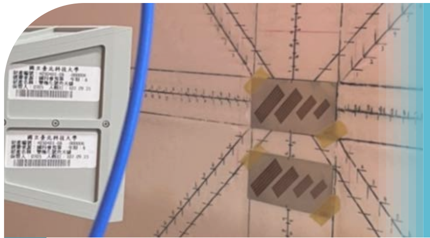
Inverse Transformation of A





Summary

- Row space and nullspace are orthogonal complement
- Column space and left nullspace are orthogonal complement
- Every vector x has a piece in the row space and a piece in the nullspace
- Only a part of x is transformed to the range (which part?)
- Can we fully invert the outcome? What is the condition that we can invert the outcome?
- If $\mathbf{A}^{-1}$ exists, it can recover all the vectors because $\mathbf{A}^{-1}(\mathbf{A}\mathrm{x}) = \mathbf{A}^{-1}\mathrm{b} = \mathrm{x}$
- When $\mathbf{A}^{-1}$ fails to exist, the pseudoinverse ($\mathbf{A}^{+}$) recovers only partial x in the row space ($\mathbf{A}^{+}(\mathbf{A}\mathrm{x}) = \mathrm{x}_r$)
- $\mathbf{A}^{+}$ is to invert what is invertible (which part?). For the left nullspace, $\mathbf{A}^{+}$ cannot do anything



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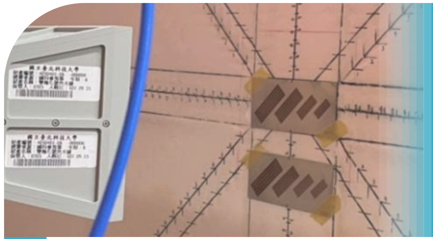
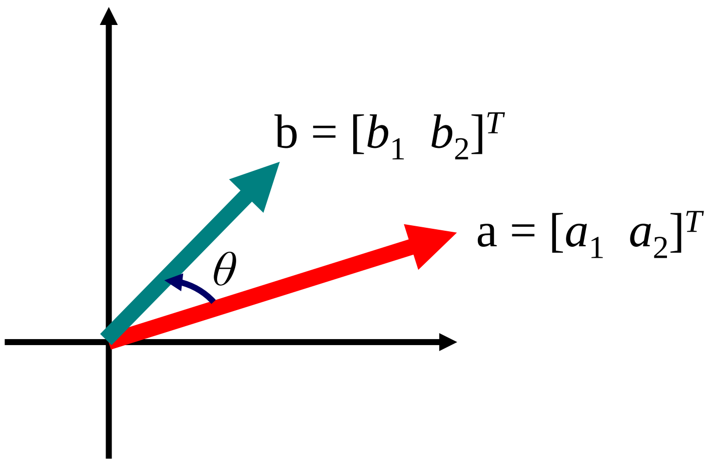
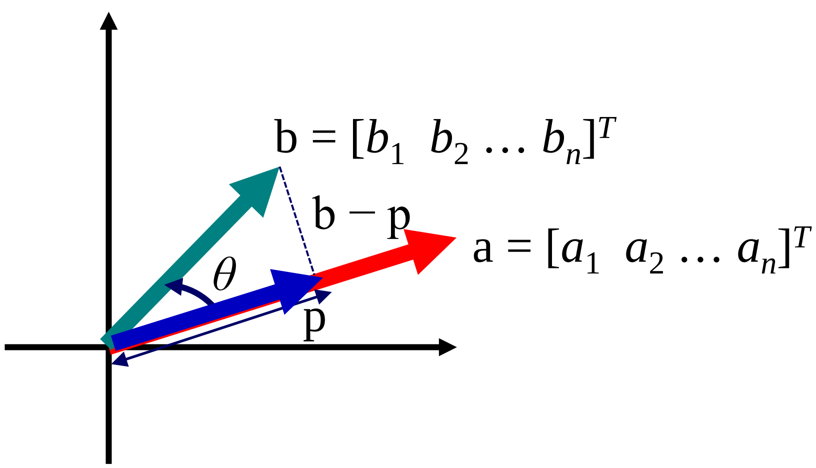


Projection of a Vector



$$\cos\theta = \frac{a^T b}{\|a\|\|b\|}$$

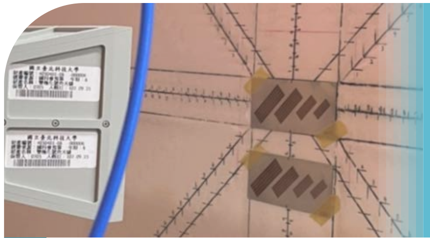


$$p = \hat{x} a$$

$$(b - p) \perp a \rightarrow a^T (b - \hat{x} a) = 0$$

$$\hat{x} =$$

$$p =$$



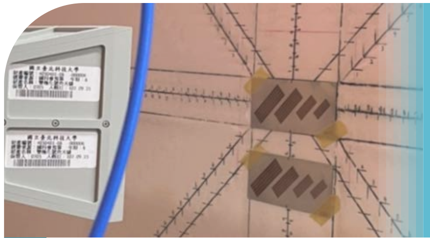
Projections of Rank One

- Now, let's apply the concept in Ch. 2.5: linear transformation
- The projection is also a linear transformation. So, what is the transformation matrix?
- The projection of b onto the line through O and a is

$$\mathrm{p} = \hat{x}\mathrm{a} = \mathrm{a}\frac{\mathrm{a}^T\mathrm{b}}{\mathrm{a}^T\mathrm{a}} = \boxed{\frac{\mathrm{a}\mathrm{a}^T}{\mathrm{a}^T\mathrm{a}}}\mathrm{b}$$

Projection matrix, **P**

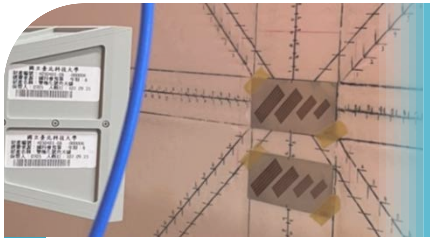
- Example: the matrix that projects any vector onto the line through $\mathrm{a} = [1\ \ 1\ \ 1]^T$:



Features about Projection Matrix

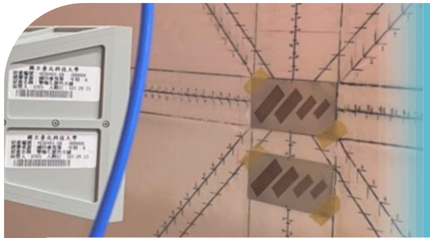
$$\mathbf{P} = \frac{\mathrm{a}\mathrm{a}^T}{\mathrm{a}^T\mathrm{a}}$$

- $\mathbf{P}^T = \mathbf{P}$. That is, $\mathbf{P}$ is always a symmetric matrix. Why?
- $\mathbf{P}^2 = \mathbf{P}$
- What vector constructs the column space?
- Rank of $\mathbf{P}$ =
- What vectors construct the nullspace?
- Can projection be inverted? Does $\mathbf{P}^{-1}$ exist?
- When a is multiplied by a scalar, will $\mathbf{P}$ stay the same?
- Projection matrix onto the θ-direction line ($\mathrm{a} = [\cos\theta \;\; \sin\theta]^T$):



An Example

- Find the matrix $\mathbf{P}$ that projects every vector b in R^3 onto the line in the direction of $\mathrm{a} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$
- What are the column space and nullspace of $\mathbf{P}$? Describe them geometrically and also give a basis for each space
- Are the vectors in the basis of the nullspace orthogonal?



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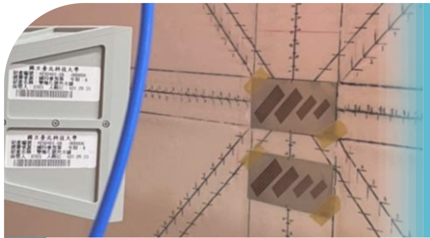
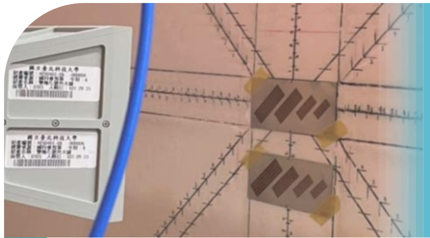


Unsolvable $\mathbf{A}x = b$

$$\mathbf{A}x = b$$

- To make $\mathbf{A}x = b$ be solvable, b must be in C($\mathbf{A}$)
- When the number of equations is greater than that of unknowns, b is difficult to be in C($\mathbf{A}$)
- Find an $\mathbf{A}x$ that is as close as possible to b

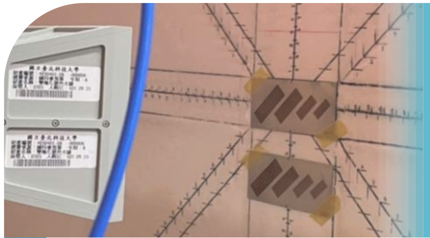


1-D Special Case

- Let's begin by a 1-D case:

$$3x = b_1$$
$$4x = b_2$$
$$5x = b_3$$

- Does a solution exist?
- If not, how do we find x that minimizes ***<u>the sum of squared errors</u>***?



1-D Special Case

- Now, considering general cases (but still 1-D):

$$E^2 = (a_1 x - b_1)^2 + \cdots + (a_m x - b_m)^2 = \|\mathrm{a}x - \mathrm{b}\|^2$$

- Set the derivative to zero:

$$a_1(a_1\hat{x} - b_1) + \cdots + a_m(a_m\hat{x} - b_m) = \mathrm{a}^T(\mathrm{a}\hat{x} - \mathrm{b}) = 0$$

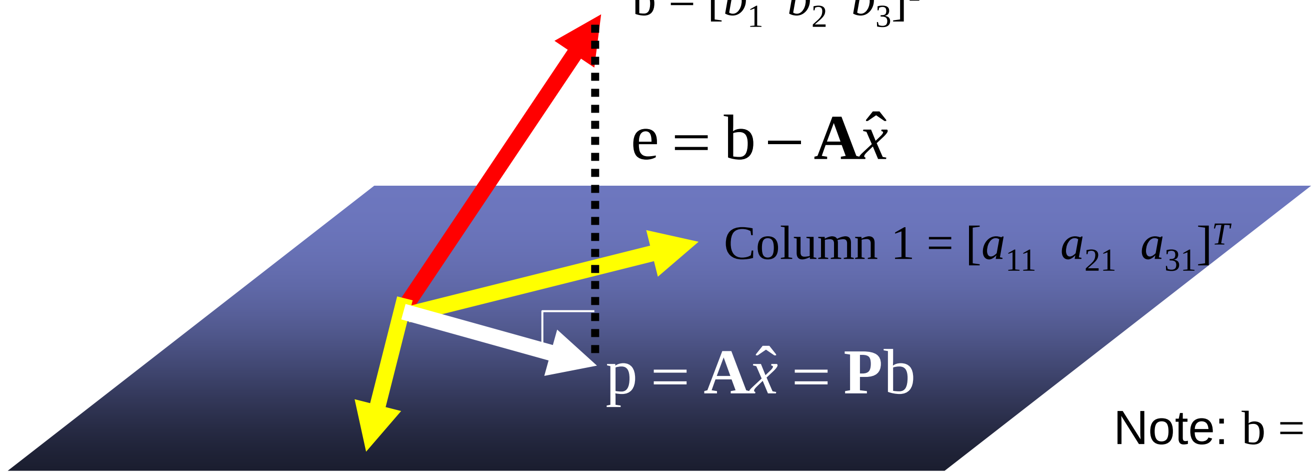
- The least squares solution to $\mathrm{a}x = \mathrm{b}$ is $\hat{x} = \dfrac{\mathrm{a}^T\mathrm{b}}{\mathrm{a}^T\mathrm{a}}$
- Have we seen this expression before? Can we compare this solution with the result in Ch. 3.2?

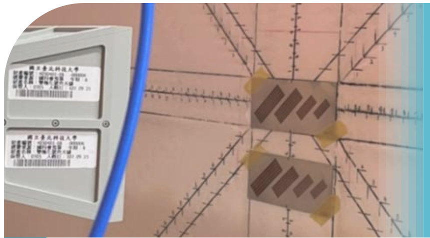
Projection onto Column Space

Find $\hat{x}$ that minimizes $E^2 \equiv$

Find a point p in C($\mathbf{A}$) that is closest to b

Instead of solving $\mathbf{A}\mathrm{x} = \mathrm{b}$, we solve $\mathbf{A}\hat{x} = \mathrm{p}$





Projection onto Column Space

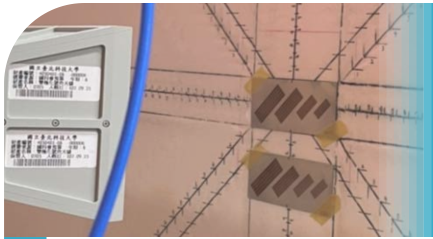
- The error $\mathrm{b} - \mathbf{A}\hat{x}$ must be perpendicular to the C($\mathbf{A}$)
- The vector perpendicular to the C($\mathbf{A}$) must be in the left nullspace:

$$\mathrm{y}^T\mathbf{A} = 0 \text{ or } \mathbf{A}^T\mathrm{y} = 0$$

$$\mathbf{A}^T\left(\mathrm{b} - \mathbf{A}\hat{x}\right) = 0$$

Normal equation!

$$\mathbf{A}^T\mathbf{A}\hat{x} = \mathbf{A}^T\mathrm{b}$$



Highlight of Solving $\mathbf{A}x = b$

$$\mathbf{A}x = b$$

Solvable:

b is in the column space of $\mathbf{A}$ (and $\mathbf{A}$ is invertible)

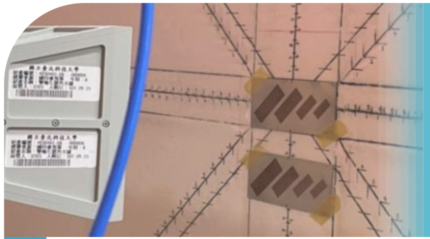
$$\mathbf{A}^{-1}\mathbf{A}x = \mathbf{A}^{-1}b$$

Unsolvable:

b is NOT in the column space of $\mathbf{A}$, so we project b onto the C($\mathbf{A}$)

$$\mathbf{A}^{T}\mathbf{A}\hat{x} = \mathbf{A}^{T}b$$

- These computations require that $\mathbf{A}$ has independent columns and a nullspace of R^0 (so $\mathbf{A}^{-1}$ and $(\mathbf{A}^T\mathbf{A})^{-1}$ exist)
- What if $\mathbf{A}$ has dependent columns? What if $\mathbf{A}$ cannot be inverted? The explanations will be given in Chapter 6



More about Projection Matrix

$$\mathbf{A}^T\mathbf{A}\hat{x} = \mathbf{A}^T\mathrm{b} \quad \Rightarrow \quad \hat{x} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T\mathrm{b}$$

- Recall the projection of b onto the C($\mathbf{A}$):

$$\mathrm{p} = \mathbf{P}\mathrm{b} \ (= \mathbf{A}\hat{x})$$

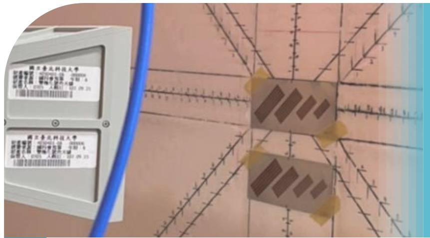
If $\mathbf{A}$ is just a vector:

$$\mathbf{P} = \frac{\mathrm{a}\mathrm{a}^T}{\mathrm{a}^T\mathrm{a}}$$

When $\mathbf{A}$ is a matrix:

$$\mathbf{P} = \mathbf{A}\left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T$$

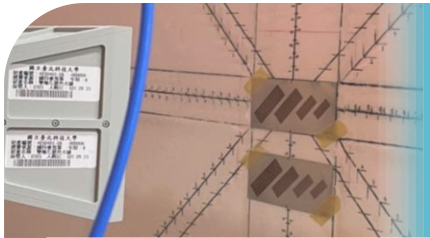
- Two special cases:
  - b is already in the C($\mathbf{A}$):
  - b is perpendicular to the C($\mathbf{A}$), namely, b is in the left nullspace:



More about Projection Matrix

$$\mathbf{P} = \mathbf{A}\left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T$$

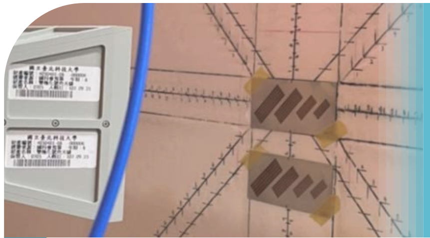
- Characteristics about $\mathbf{A}^T\mathbf{A}$:
  - Symmetric
  - $\mathbf{A}^T\mathbf{A}$ has the same nullspace as $\mathbf{A}$
  - If $\mathbf{A}$ has linearly independent columns, $\mathbf{A}^T\mathbf{A}$ is square and invertible
  - $\hat{x}$ can be solved when $\mathbf{A}$ has full column rank (n independent columns)
- When b is projected, it is split into 2 components
  - $\mathbf{P}\mathrm{b}$ in the column space
  - $\mathrm{b} - \mathbf{P}\mathrm{b}$ in the left nullspace (orthogonal component), so $\mathbf{I} - \mathbf{P}$ is also a projection matrix



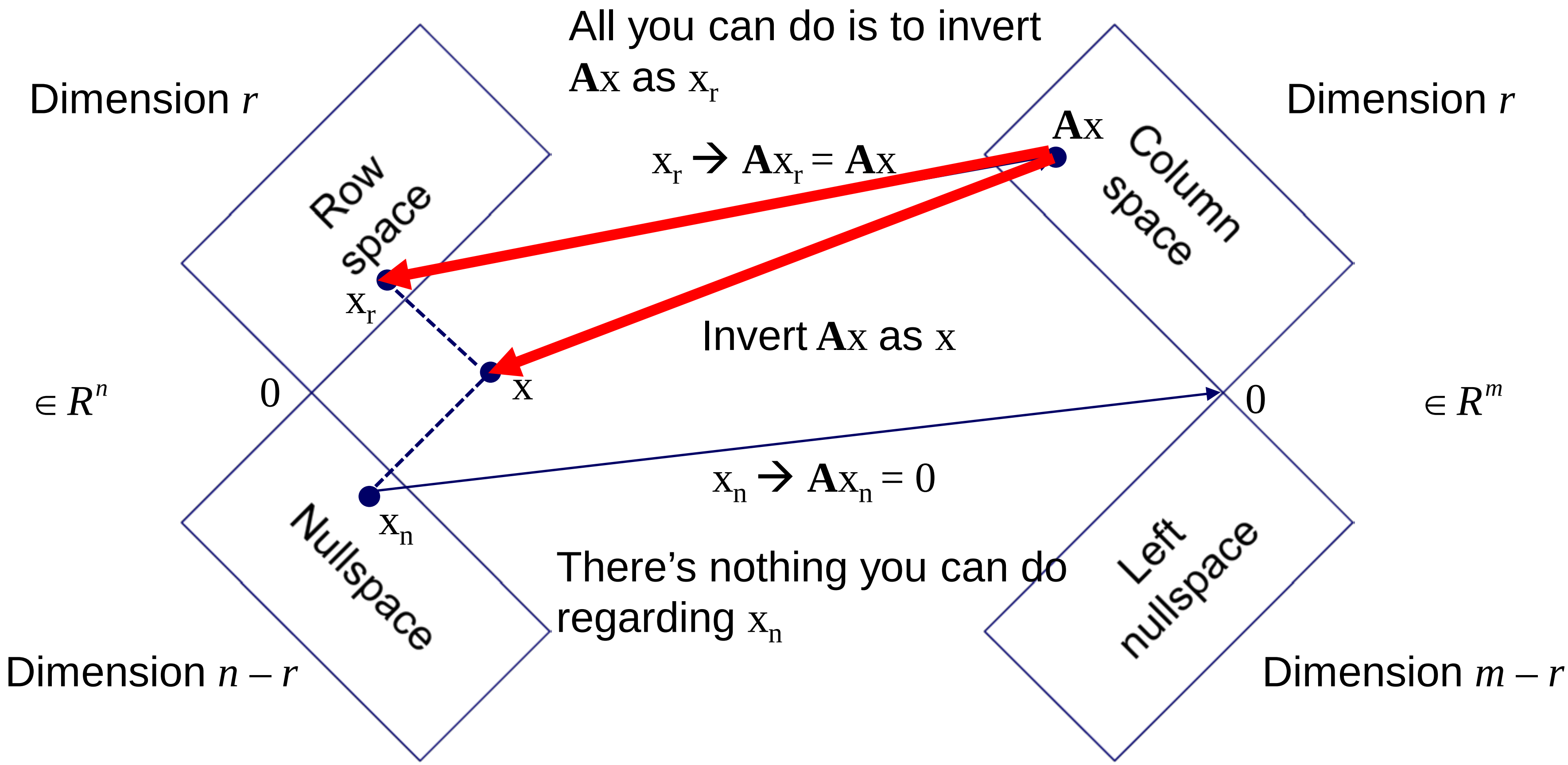
More about Projection Matrix

$$\mathbf{P} = \mathbf{A}\left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T$$

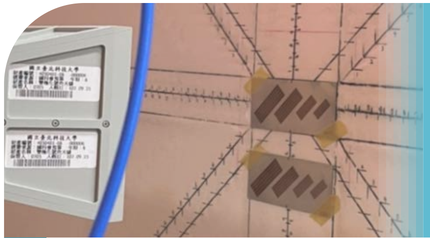
- $\mathbf{P}^T = \mathbf{P}$
- $\mathbf{P}^2 = \mathbf{P}$
- $\mathbf{P}$ is the projection matrix that projects a vector onto the column space of $\mathbf{A}$
- What is the projection matrix that projects a vector onto the row space of $\mathbf{A}$?
- What is the projection matrix that projects a vector onto the nullspace of $\mathbf{A}$?
- What is the projection matrix that projects a vector onto the left nullspace of $\mathbf{A}$?



An Application of P



Invert **A**x as x_r is exactly the "pseudoinverse" of **A**, namely, $\mathbf{A}^+$



An Application of P

- Pseudoinverse: Recover $\mathbf{A}\mathrm{x}$ as x_r

$$\mathbf{A}^{+}\left(\mathbf{A}\mathrm{x}\right)=\mathrm{x}_{r}$$

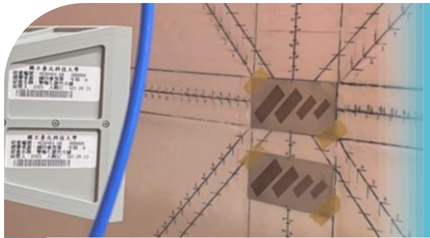
- x_r is the projection of x onto the row space

$$\mathbf{A}^{+}\left(\mathbf{A}\mathrm{x}\right)=\left[\mathbf{A}^{T}\left(\mathbf{A}\mathbf{A}^{T}\right)^{-1}\mathbf{A}\right]\mathrm{x}$$

The formula of $\mathbf{A}^{+}$:

$$\mathbf{A}^{+}=\mathbf{A}^{T}\left(\mathbf{A}\mathbf{A}^{T}\right)^{-1}$$

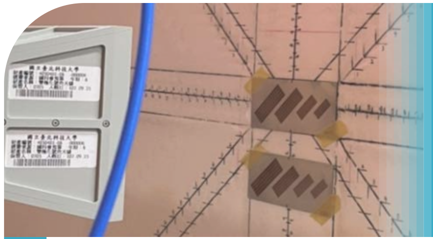
- $\mathbf{A}\mathbf{A}^{+}=\mathbf{I}$



Continue the Previous Example

For a system $\mathbf{A}\mathrm{x} = \mathrm{b}$ where $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}$, $\mathrm{x} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$, and $\mathrm{b} = \begin{bmatrix} 9 \\ 18 \end{bmatrix}$,

- find the bases for the row space and the nullspace of $\mathbf{A}$
- Split x into a row-space component x_r and a nullspace component x_n
- Find the pseudoinverse of $\mathbf{A}$
- Use the pseudoinverse to recover the row-space component x_r



Example of Least Squares

- Several observations are sampled over time:
  $(t_1, b_1), (t_2, b_2), \ldots, (t_m, b_m)$

- Let's fit the observation to a model $C + Dt = b$ so that we can predict the incoming data

$$C + Dt_i = b_i$$

$$\Rightarrow \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \Rightarrow$$

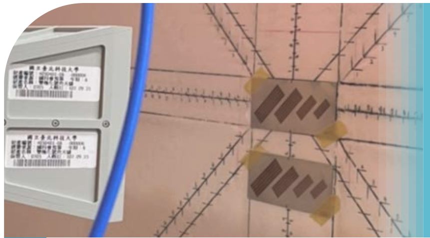
It is a problem of $\mathbf{A}\mathrm{x} = \mathrm{b}$

For example,

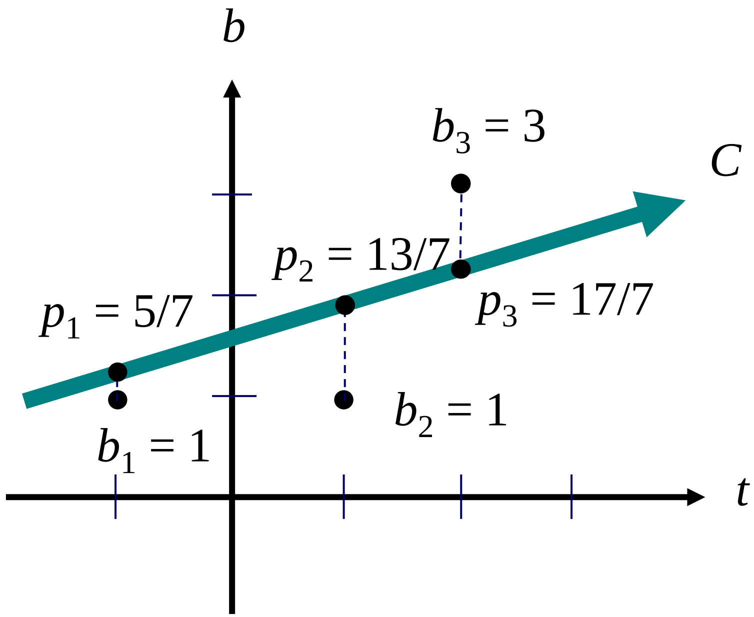
$(t_1, b_1) = (-1, 1)$

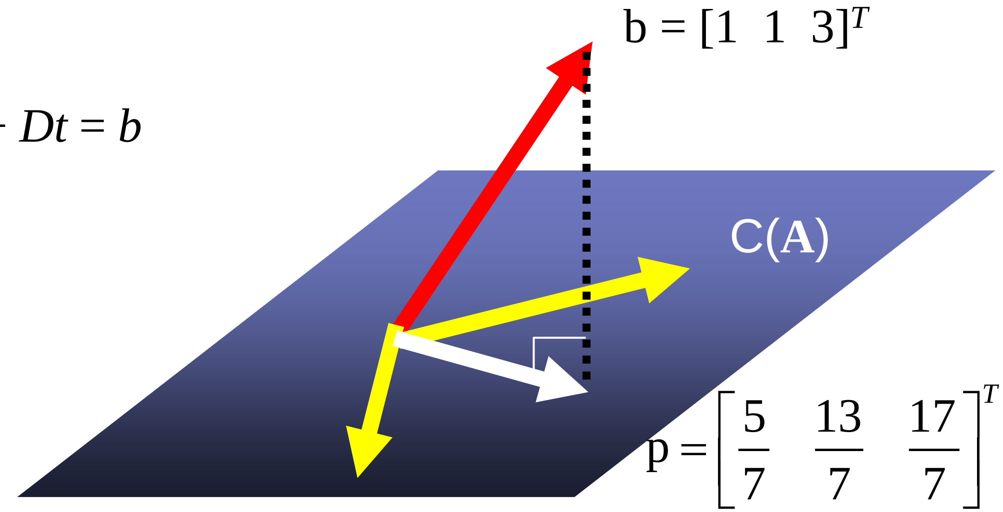
$(t_2, b_2) = (1, 1)$

$(t_3, b_3) = (2, 3)$

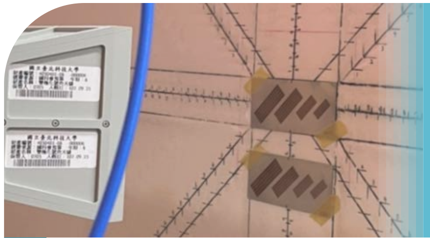


Two Explanations





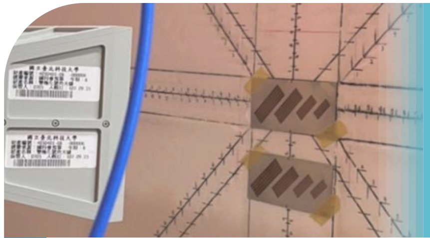
- $(-1, 1)$, $(1, 1)$, and $(2, 3)$ are not on the line
- Least-squares replaces them by $(-1, p_1)$, $(1, p_2)$, and $(2, p_3)$
- $\mathrm{b} = [1 \;\; 1 \;\; 3]^T$ is not on the plane
- Least-squares replaces b by p which lies on the plane
- Check $\mathrm{e} = \mathrm{b} - \mathrm{p}$ and both columns are orthogonal



Multiple Regression

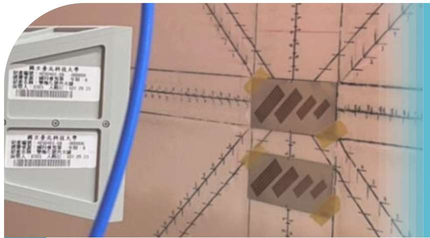
Observation	Response Y	Input factor X_1	Input factor X_2	$\cdots$	Input factor X_p
1	y_1	x_{11}	x_{12}		x_{1p}
2	y_2	x_{21}	x_{22}		x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$
n	y_n	x_{n1}	x_{n2}		x_{np}

- Regression model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + e$
- Goal: Estimate β_i based on n sets of observation
- How do we model the problem into $\mathbf{A}\mathrm{x} = \mathrm{b}$?
- How do we determine $\hat{\beta}_i$?



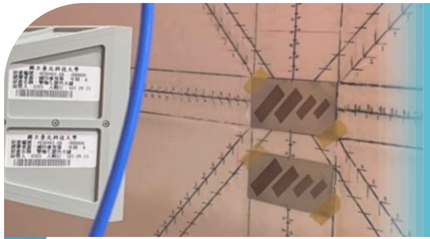
Prediction for Revenue

Movie title	Production budget ($M)	Trailer Views (M)	IMDB score	Lead Actors' IG Followers (M)	Opening Weekend Revenue ($M)
Oppenheimer	100	72	8.3	2 + 58	82
Mission: Impossible - Dead Reckoning Part One	290	31	7.7	13	55
Deadpool & Wolverine	200	35	7.8	54 + 35	211
Dune: Part Two	190	29	8.5	19 + 182	83
Joker: Folie à Deux	190	37	5.3	58	



Agenda

1. Orthogonality of Subspaces
2. Projection onto 1-D Subspaces
3. Least Squares Approximations
4. **Orthonormal Bases**
5. Gram-Schmidt Process



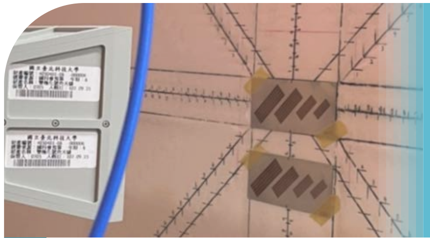
Orthonormal Bases

- For a given space or subspace of R^n, there are infinite numbers of bases
- Each basis consists of n vectors
- What basis is the best you can get?

Orthonormal basis $q_1, \ldots, q_n$

- The vectors $q_1, \ldots, q_n$ are ***orthonormal*** if

$$q_i^T q_j = \begin{cases} 0, & \text{if } i \neq j \quad \textit{(orthogonality)} \\ 1, & \text{if } i = j \quad \textit{(normalization)} \end{cases}$$



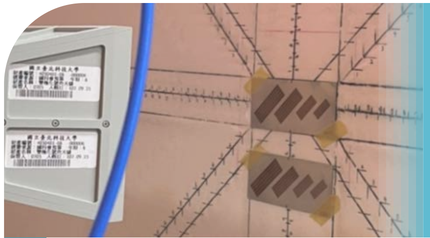
Orthogonal Matrix

Orthogonal matrix

- A ***square*** matrix with orthonormal columns
- The most important property:

$$\mathbf{Q}^T\mathbf{Q} = \begin{bmatrix} - & \mathrm{q}_1^T & - \\ - & \mathrm{q}_2^T & - \\ & \vdots & \\ - & \mathrm{q}_n^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ \mathrm{q}_1 & \mathrm{q}_2 & \cdots & \mathrm{q}_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = \mathbf{I}$$

- That is, $\mathbf{Q}^T = \mathbf{Q}^{-1}$
- $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$



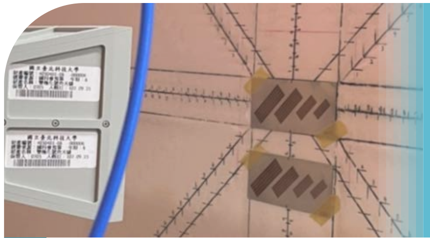
Operational Schemes of Q

- An orthogonal matrix performs:
  - Rotation
  - Reflection
  - A product of rotation and reflection

- Example:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

- A permutation matrix is a special case of the reflection

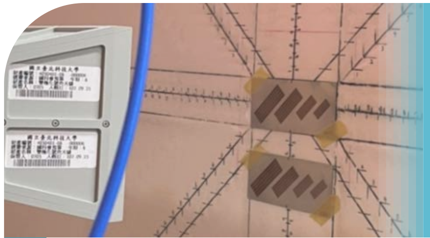


Generating Orthogonal Matrices

- Householder reflections:
  1. Start with a unit vector u
  2. Create $\mathbf{H} = \mathbf{I} - 2\mathrm{uu}^T$
  3. $\mathbf{H}$ will be an orthogonal and symmetric matrix

- Hadamard matrices (before normalization):

$$\mathbf{H}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad \mathbf{H}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{H}_2 & \mathbf{H}_2 \\ \mathbf{H}_2 & -\mathbf{H}_2 \end{bmatrix}$$



Generating Orthogonal Matrices

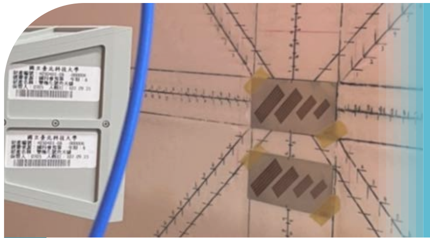
- Wavelet matrices (before normalization):

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & -1 & 0 & -1 \end{bmatrix}$$

$$\mathbf{W}_8 = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & -1 \end{bmatrix}$$

The wavelet squeezes down and rescale

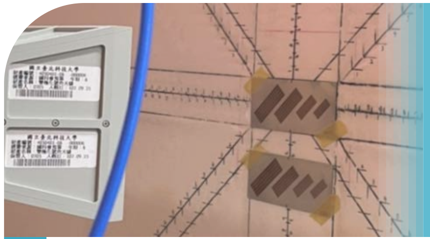


Features of $\mathbf{Q}$

- Multiplication by an orthogonal matrix preserves length:

- Multiplication by an orthogonal matrix preserves angle:

- Transform by $\mathbf{Q}$ means space rotation or reflection. They preserve inner product and length
- The rows of $\mathbf{Q}$ are also orthonormal. Why?
- $\mathbf{Q}$ plays an important role in Chapters 5 and 6, where $\mathbf{Q}$ is a matrix comprised of orthonormal eigenvectors



Why Are Orthonormal Bases Important?

- If a vector b can be expressed as the combination of $q_1, \ldots, q_n$,

$$b = x_1 q_1 + x_2 q_2 + \cdots + x_n q_n$$

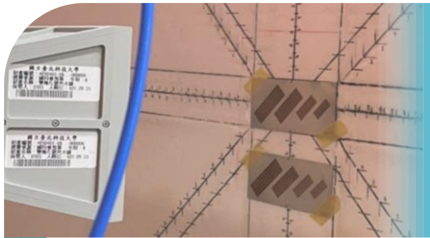
determining the coefficients (x_i) is easy

- Algebraically, multiplying both sides by q_i^T:

- The problem is identical to solving $\mathbf{Q}x = b$:

- Geometrically, x_i comes from the 1-D projection:

- From each point of view, $x_i = q_i^T b$



Rectangular Case

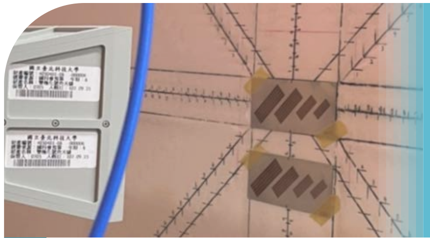
- Orthogonal matrix: it must be square
- However, rectangular matrices with orthonormal columns can also be denoted by $\mathbf{Q}$. They cannot be called "orthogonal matrices"
- Rectangular $\mathbf{Q}$ with $m > n$ → No exact solution for $\mathbf{Q}\mathrm{x} = \mathrm{b}$. How do we find the approximate $\hat{x}$ in the column space of $\mathbf{Q}$?

$$\mathbf{Q}\hat{\mathrm{x}} = \mathrm{b}$$

$$\hat{\mathrm{x}} =$$

- The least squares problem becomes easy!
- The projection matrix becomes:

$$\mathbf{P} =$$



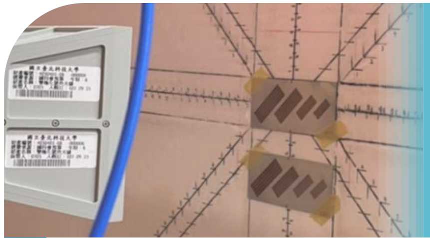
Example

- Fit the observation to a model $C + Dt = b$. This time we select observational times to have an average of zero:

$$C + Dt_i = b_i$$

$$\Rightarrow \begin{bmatrix} 1 & -5 \\ 1 & 0 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- Two columns are orthogonal
- We can project b_i separately onto each column
- How do we make observational time to have an average of zero? Centering!



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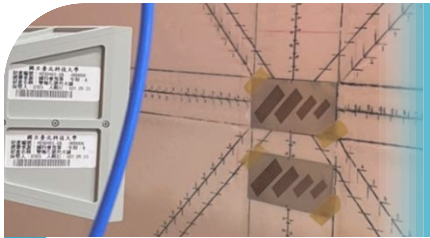


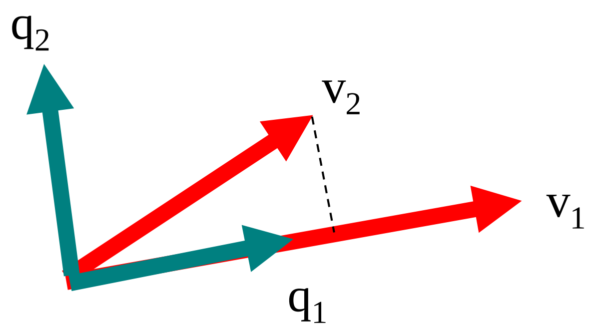
Goal of Gram-Schmidt

- To have orthogonal columns is important and powerful
- Make columns perpendicular to one another, and further generate an orthonormal basis → **Gram-Schmidt process**

Gram-Schmidt process

Given k independent vectors $v_1, v_2, \ldots, v_k$, Gram-Schmidt turns them into orthonormal vectors $q_1, q_2, \ldots, q_k$, which form the same space as that formed by $v_1, v_2, \ldots, v_k$

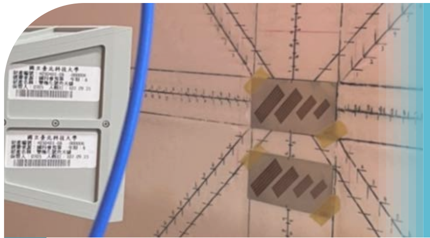
Formula



1. q_1 is selected as v_1 with normalization: $q_1 = \dfrac{v_1}{\|v_1\|}$

2. q_2 comes from v_2 that substrates off the component that belongs q_1: $q_2' = v_2 - (q_1^T v_2) q_1 \Rightarrow q_2 = \dfrac{q_2'}{\|q_2'\|}$

3. Similarly, q_3: $q_3' = v_3 - (q_1^T v_3) q_1 - (q_2^T v_3) q_2 \Rightarrow q_3 = \dfrac{q_3'}{\|q_3'\|}$



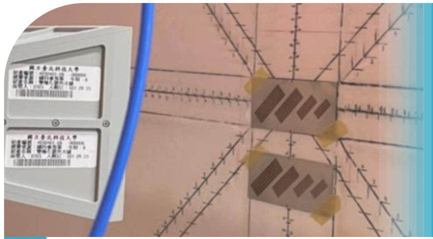
Formula

- Subtracting from every new vector where its components in the directions that are already settled

$$\mathrm{q}'_j = \mathrm{v}_j - \left(\mathrm{q}_1^T \mathrm{v}_j\right)\mathrm{q}_1 - \left(\mathrm{q}_2^T \mathrm{v}_j\right)\mathrm{q}_2 - \cdots - \left(\mathrm{q}_{j-1}^T \mathrm{v}_j\right)\mathrm{q}_{j-1} \Rightarrow \mathrm{q}_j = \frac{\mathrm{q}'_j}{\left\|\mathrm{q}'_j\right\|}$$

- Recall that "$\mathbf{I}-\mathbf{P}$ is also a projection matrix". The new vector is somewhat like

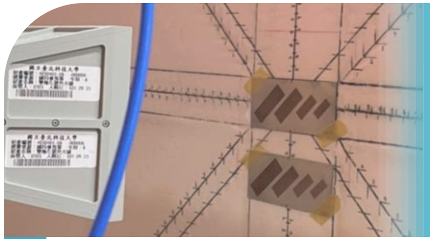
$$\begin{aligned}
\mathrm{q}'_2 &= \mathrm{v}_2 - \left(\mathrm{q}_1^T \mathrm{v}_2\right)\mathrm{q}_1 \\
&= \mathrm{v}_2 - \left(\text{the component that } \mathrm{v}_2 \text{ projects onto } \mathrm{q}_1\right) \\
&\approx \left(\mathbf{I}-\mathbf{P}\right)\mathrm{v}_2
\end{aligned}$$



Example

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

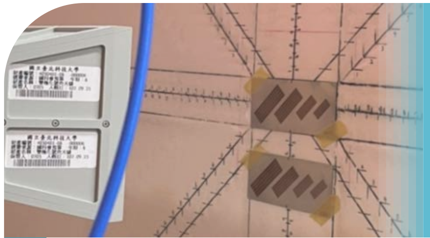
- Construct a $\mathbf{Q}$ that has the same column space as $\mathbf{A}$



Initially, Λ is just a collection of numbers.
Now our understanding advances.

Λ becomes a collection of basis, which can be determined by our preference!

All the elements have been changed, yet the new elements have the same meaning as the original one with more useful features



A = QR

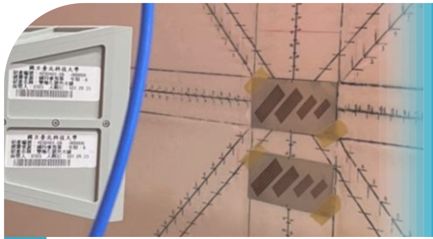
$$\mathbf{A} = \mathbf{Q}\mathbf{R}$$

- Factorization of $\mathbf{A}$ into $\mathbf{Q}$ times $\mathbf{R}$, where $\mathbf{Q}$ is formed by orthonormal columns:

$$\begin{bmatrix} & & \\ \mathrm{v}_1 & \mathrm{v}_2 & \mathrm{v}_3 \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ \mathrm{q}_1 & \mathrm{q}_2 & \mathrm{q}_3 \\ & & \end{bmatrix} \begin{bmatrix} \mathrm{q}_1^T\mathrm{v}_1 & \mathrm{q}_1^T\mathrm{v}_2 & \mathrm{q}_1^T\mathrm{v}_3 \\ & \mathrm{q}_2^T\mathrm{v}_2 & \mathrm{q}_2^T\mathrm{v}_3 \\ & & \mathrm{q}_3^T\mathrm{v}_3 \end{bmatrix}$$

Gram-Schmidt process

Upper triangular matrix



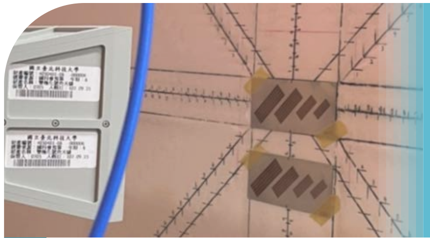
$\mathbf{A} = \mathbf{QR}$

- Every m by n matrix $\mathbf{A}$ with linearly independent columns can be factored into $\mathbf{A} = \mathbf{QR}$
- The columns of $\mathbf{Q}$ are orthonormal
- $\mathbf{R}$ is upper triangular and invertible
- When $m = n$, all matrices are square, and $\mathbf{Q}$ becomes an orthogonal matrix

- Why do we need $\mathbf{A} = \mathbf{QR}$? Recall the least squares solutions

$$\hat{x} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T\mathrm{b}$$

The solution can be obtained by only back substitution!



Hilbert Space

Function space or Hilbert space

- R^∞ with finite length
- $\mathrm{v} = (v_1, v_2, v_3, \ldots)$ with $\|\mathrm{v}\|^2 = v_1{}^2 + v_2{}^2 + v_3{}^2 + \ldots$ finite

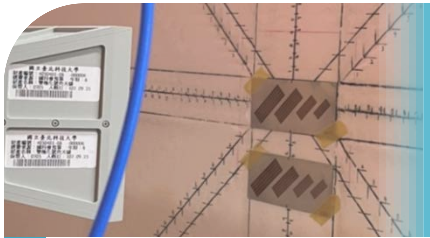
- Example: $(1, \frac{1}{2}, 1/3, \frac{1}{4}, \ldots)$ and $(1, 1, 1, 1, \ldots)$. Which is in Hilbert space?
- In Hilbert space, the properties of vectors are preserved:
  - $\mathrm{v} \perp \mathrm{w} \rightarrow \mathrm{v}^T\mathrm{w} = v_1 w_1 + v_2 w_2 + v_3 w_3 + \ldots = 0$
  - Schwarz Inequality $|\mathrm{v}^T\mathrm{w}| \leq \|\mathrm{v}\| \times \|\mathrm{w}\|$
  - Vectors can be turned into functions

Examples

- $f(x) = \sin x \ (0 \le x \le 2\pi)$
  - We have infinitely many points of $f(x)$ along the whole interval of x
  - Why is it like a vector?
  - How to find the length (or, norm,) of this vector?
  - Why is the length finite?

- $g(x) = \cos x \ (0 \le x \le 2\pi)$
  - How do we determine the ***inner product*** of two functions
  - Why are $f(x)$ and $g(x)$ orthogonal?

- Two functions are orthogonal if $f^T g = 0$ and are orthonormal after division by their lengths



Fourier Series

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \ldots$$

- To compute the coefficient, say, b_1:

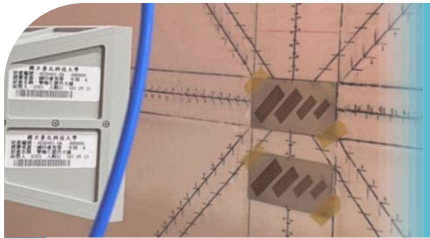
$$\int_0^{2\pi} f(x) \sin x dx =$$

$$b_1 = \frac{\int_0^{2\pi} f(x) \sin x dx}{\int_0^{2\pi} \sin x \times \sin x dx} = \frac{\langle f, \sin x \rangle}{\langle \sin x, \sin x \rangle}$$

Remember

$$\hat{x} = \frac{\mathrm{a}^T \mathrm{b}}{\mathrm{a}^T \mathrm{a}} \text{ in } \mathrm{p} = \hat{x}\mathrm{a}?$$

We are projecting f onto $\sin x$!



Legendre Polynomials

- Why is Fourier series beautiful? The basis vectors consisting of $\{1, \cos x, \sin x, \cos 2x, \sin 2x, \ldots\}$ are orthogonal!
- How about polynomial functions? The basis is $\{1, x, x^2, x^3, \ldots\}$

$$f\left(x\right) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots$$

- This is not a orthogonal basis!
- This brings additional trouble in solving $\mathbf{A}^T \mathbf{A} \hat{x} = \mathbf{A}^T \mathrm{b}$
- How do we make the basis of polynomial functions $\{1, x, x^2, x^3, \ldots\}$ orthogonal?

Legendre Polynomials

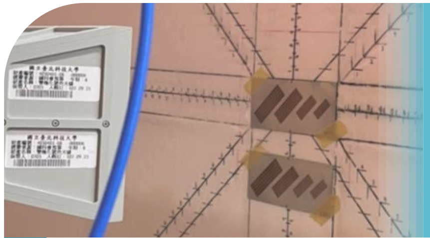
Gram-Schmidt!

- Considering $-1 \le x \le 1$:

The original basis	The new basis
1	
x	
x^2	
x^3	
X^4	

- $\langle 1, x \rangle =$
- $\langle x, x^2 \rangle =$
- $\langle 1, x^2 \rangle =$

$$\mathrm{v}_3 = x^2 - \frac{\langle 1, x^2 \rangle}{\langle 1, 1 \rangle} 1 - \frac{\langle x, x^2 \rangle}{\langle x, x \rangle} x =$$



Examples

- Approximate $y = x^5$ by $C + Dx$ for $0 \le x \le 1$:

Method 1: Solve

$$\begin{bmatrix} 1 & x \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = x^5$$

Method 2: Minimize

$$E^2 = \int_0^1 \left(x^5 - C - Dx \right)^2 dx$$

wrt C and D

Method 3: Apply Gram-Schemidt to replace x

$$\mathrm{v}_2 = x - \frac{\langle 1, x \rangle}{\langle 1, 1 \rangle} 1$$

and project x^5 onto the two basis vectors